

Dr. James T. McLeskey Jr. & Dr. Zachary Cullingsworth

Megan Bost, Grayson Bush, Davis Clemmons, & Nate DeLong

Letter of Transmittal

December 13, 2023

Dear Dr. James T. McLeskey Jr. and Dr. Zachary Cullingsworth,

This letter is to inform you of the different sections of this design project. The first part is the design report which includes the research and background about the chosen design and the plan used to construct it. The design report will be accompanied by a group powerpoint presentation explaining the contents of the design report. The next part is a working prototype of the design. And the final piece is a poster to be presented at the final conclusion of the Capstone Senior Design class. All of these parts were worked on collaboratively and combined as a final product together as a group. All of the parts mentioned above should be completed and available to you by the conclusion of the spring semester on May 13th, 2024.

Sincerely,

Megan Bost, Grayson Bush, Davis Clemmons, & Nate DeLong



RANDOLPH-MACON

C O L L E G E

Department of Physics, Engineering, and
Astrophysics

Capstone Design Report for the Minto Wheel Heat Engine

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Executive Summary

The generation of energy and power are two things that continue to be studied and improved upon even today. The need to produce energy cleanly and efficiently is a driving force for many energy-producing companies. The Minto Wheel is a heat engine that uses a difference in temperature to create mechanical work which can be converted into power. This design uses solar energy and water to provide energy for the Minto Wheel to successfully generate work. The goal of this design is to construct a working wheel with low cost materials. The medium being used is propane which has a low boiling point and is cost effective. The temperature of the solar-heated water will also be taken into consideration to ensure the low boiling liquid reaches an appropriate temperature. The construction of the Minto Wheel design will start on February 12, 2024. This design aims to reach a certain goal of efficiency and build a working prototype of the Minto Wheel by May 6, 2024. Equipment and parts for assembly will be provided by the Department of Physics, Engineering, and Astrophysics at Randolph-Macon College, along with parts and equipment from outside sources.

Acknowledgements

We would like to thank Dr. James T. McLeskey Jr. and Dr. Zachary Cullingsworth for their continued advice, support, and guidance on this project. Along with their guidance, we appreciate their continued belief in the group.

We further would like to thank the Department of Engineering, Physics, and Astrophysics at Randolph-Macon College for providing the resources to make this project possible.

Thank you for your continued support,

Megan Bost, Grayson Bush, Davis Clemmons, & Nate DeLong

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Introduction and Background

The need for electrical energy continues to be apparent in today's society. That need grows in demand as countries around the world continue to develop and grow. The issue with electrical energy is generating it in a clean and efficient manner. There are several ways that energy can be generated, but issues continue to appear with each of these ways. Coal and natural gas are the most commonly used, but they are nonrenewable and pollute the planet's atmosphere when used. Wind and solar are clean, but not super efficient and are more costly. These considerations put the world at a crossroads of picking clean energy or efficient energy. The Minto Wheel is a tool we will be considering for the production of clean energy.

In 2022 the world consumed 25,530 terawatt-hours of electricity (Global Electricity Consumption, 2022). That amount is larger than 2021 which was larger than 2020 and so on. In the same year, in generating that 25,530 terawatt-hours of electricity, 1,539 million metric tons of Carbon Dioxide were produced in the United States alone (Energy Information Administration, 2023). The most common ways of generating electricity in the US, using coal and natural gas, account for 55% and 43% of the CO₂ respectively (EIA - independent statistics and analysis, 2023). All of this put together clearly shows that America's and the world's need for electricity has some harmful effects on the planet.

There are already several forms of clean energy currently in use, but there is one form of energy that hasn't been developed and used as much as the others, and that is solar energy. The Minto Wheel can run 24/7, through sunlight and artificial heat, and produce energy without polluting the environment. It relies on the ideas of a low-boiling liquid and temperature gradient to spin. It can be created out of basic materials and can be made in many sizes so it is adaptable to the needs and resources of where it would be implemented.

The wheel consists of a base, sealed containers with low-boiling liquid, a heat source, and the mechanism used to harness the energy from it. The full containers are attached to the spinning base opposite from an empty container. There is a heat source, usually a hot water bath, at the bottom of the wheel. This is used to heat the low-boiling liquid to create a vapor pressure in the container. Once the pressure is high enough, the vapor in the container begins to rise. As the warm vapor reaches the cooler top container, it begins to condense causing the top container to get heavier and want to fall towards the bottom, where the liquid will be reheated. This process generates an enormous amount of torque which in turn can be used to generate electricity. The process as a whole can be seen in Figure 1.

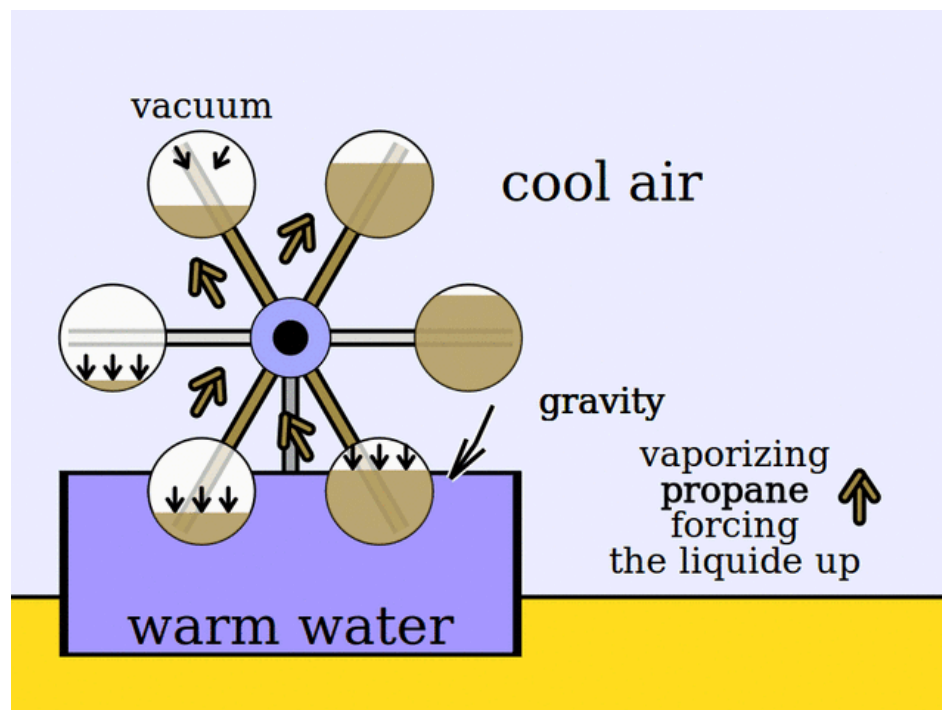


Figure 1: The overview of the Minto Wheel process (Frey, 2016).

The goal of this project is to create a wheel that does as described while using low-cost materials. This can help create clean, reusable power for people with little resources. The final

product will be a prototype or recreated physical design of the wheel using the resources provided. Some design constraints that have been considered throughout the creation and construction of the design are resources or materials available within a certain price range and the amount of time given to build a working design.

Identifying Opportunities:

The idea for this project and the research associated with it came from the ENGR 450-01 class and Dr. McLeskey. In the beginning of the year each student in ENGR 450-01 was tasked with developing three ideas for a Capstone project. Once all the ideas from the class were brought forward, the class then decided which ideas were most realistic to accomplish in the time span required. One of this project's group members, Megan Bost, brought in the idea of a water powered energy wheel. This idea was taken and then tweaked by Dr. McLeskey to pivot to construct a Minto wheel which is the current focus of the group. This idea is important to the group because the need for energy in this world is only increasing. It is important to find a sustainable way to continue producing energy. This project allows the development of a cheap, clean, and reusable way to do that. Other ideas this group considered to generate clean energy were the water powered energy wheel mentioned above, a small scale wind turbine, and a solar powered battery. However, given the time frame and skill set of the group, the Minto wheel was chosen.

Developing Requirements:

There are a few design constraints that were placed on the project and needed to be considered when coming up with the final design. The first of these constraints is the time frame in which the project must be completed, and that a working prototype should be produced. The idea for the design was decided and a proposal was drafted by October 16, 2023. The design is to

be completed and presented on by May 13, 2024. This is a very specific deadline in which the project must be completed. One of the requirements of the project is to design a working prototype, which was ultimately considered to create a reasonable design that is challenging but can still be completed within the time frame. The other major constraint that was considered was the resources and materials that we can use to produce the prototype. The group was required to stay within a certain budget, and the cost of materials would need to be considered to stay within a certain price range and afford all of the needed materials.

Literature Review

As our population and society grow and more of the world is modernized, the need for power increases. There are several available options to create power, but most of them are inefficient or nonrenewable (Krishnan, 2015). In 1976, A man named Wallace Minto developed a wheel that solves both of those issues. The Minto Wheel capitalizes off of mechanical and thermodynamic principles such as gravity and boiling points. The wheel functions by having a low boiling point substance in closed containers which are mounted around the outside of a wheel. On the bottom of the wheel, the containers are heated up using some sort of heat source such as a hot water bath. This causes an increase in vapor pressure which drives the working fluid to the top of the wheel causing the cooler, less pressurized containers full of now condensed liquid at the top to fall and be reheated. This cycle continues to spin the wheel at a very low speed and the large torque generated from it can be used to generate power, (Lindsley, Wally Minto's Wonder Wheel).

The first example of this wheel in action was published by Mother Earth News in 1976. They published an article describing how the wheel works along with some quotes from Minto himself and a picture of him operating his wheel. This picture of Minto next to his wheel can be seen below in Figure 2. Minto was quoted talking about how the wheel could be scaled to much larger sizes and generate more horsepower, (Mother Earth News). Minto's dream ended up turning into a reality as his wheel did not die with him. Although his original wheel was about 6 feet in diameter and used Freon as the low boiling temperature liquid, there have been several adaptations to his design changing both the size and low boiling point liquid, (Mother Earth News).

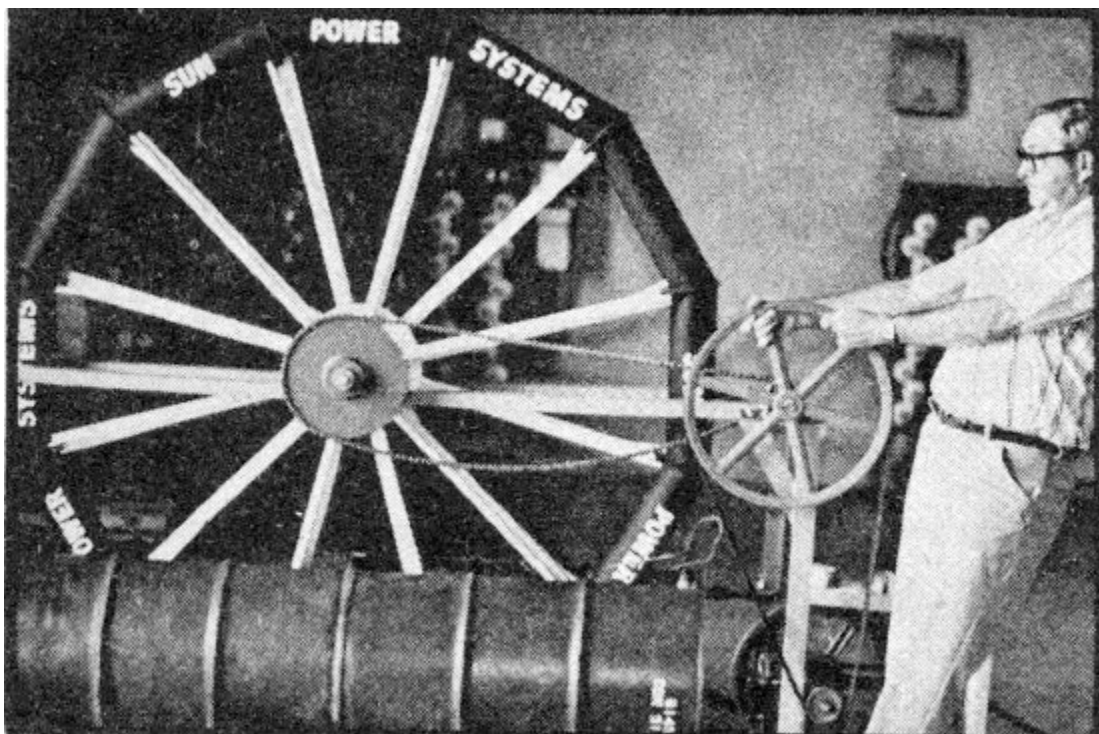


Figure 2: Wally Minto standing next to his invention, the Minto Wheel, (Mother Earth News).

There have been several variations of the Minto Wheel patented using different heating methods and different types of engines, (Wikimedia Foundation, 2023). While the wheel is

named after Minto, the idea for the wheel had been created long before his work in the 1970s. In 1881, two sets of inventors, the Iske brothers and Israel L. Landis, submitted patents for thermodynamic engines that worked in the same way as the Minto wheel. The Iske brothers' patent only differed from the Minto wheel in the working fluid that was used and the heat source that provided the energy. Landis' patent differed from the others in the fact that it created power through oscillating tanks as opposed to the spinning wheel that both the Iske brothers and Minto created, (Academic Accelerator, 2023).

A derivation of Landis's patent is the "Thermoscopic Balance " mentioned in the 1987 Scientific American journal. This balance was an oscillating desk toy in which energy was harnessed from the sun to drive the working fluid from one tank to the other causing it to oscillate back and forth periodically, (Hopkins, 1887). In 1941 a patent was granted to Karl Jauch for a device that would eventually be known as the "German Puja Clock." This clock used a spinning wheel to act as the winding key for the clock. The thermodynamic principles used to make the wheel spin are the same as those seen in the Minto wheel, (Album: Puja, 2023). It is not clear exactly where Minto got the idea for his wheel but it is likely from the Iske brothers patent because they are constructed so similarly.

Another variation of Minto's wheel is the Drinking Bird, patented in the United States by Miles V. Sullivan in 1945, (Drinking Bird, Wikipedia, 2021). The drinking bird is a heat engine that works similarly to the Minto wheel in that it uses a temperature difference to convert heat energy into mechanical work. This design consists of a glass tube with glass bulbs connected at each end. To operate, the head of the "bird" must thoroughly dip into a container of room temperature water. The cooling of the head of the bird condenses the vapor within to a liquid which causes a pressure drop. The pressure drop pulls the liquid from the bottom bulb up the

tube into the head and the weight of the fluid in the head causes the bird to tilt and re-dip into the water, while simultaneously letting vapor from the bottom bulb into the top bulb and equalizing the pressure. The liquid falls back to the bottom bulb and the bird returns to the starting position, to repeat the process all over. This device is not a perpetual motion device because, without the water present, the bird will eventually stop motion. This example is an oscillating version of the Minto Wheel. Instead of making a complete revolution, the drinking bird tips back and forth. Similarly to the wheel, the drinking bird moves due to differential pressure caused by a temperature change. An image of the drinking bird can be seen below in Figure 3.

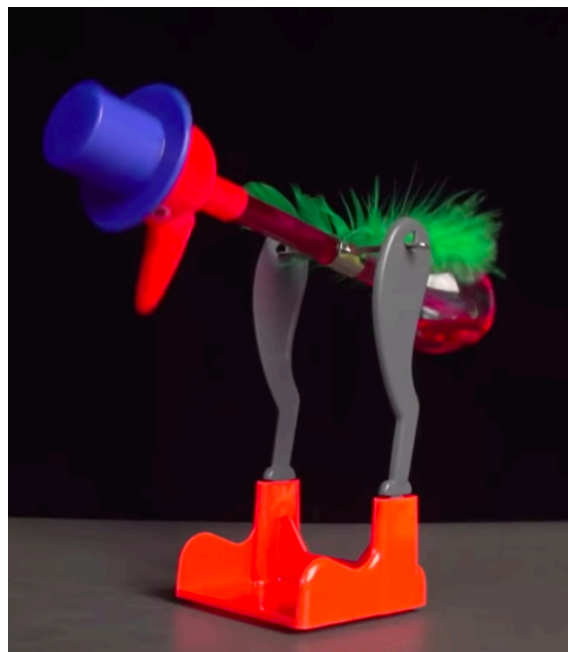


Figure 3: The drinking bird design by Miles Sullivan, (Drinking Bird, Wikipedia, 2021).

With his work in the 1970's, Minto made improvements to the wheel by experimenting with different working fluids and employing solar power to make the engine sustainable for long periods, (Krishnan, 2015). There have been many attempts to construct the Minto Wheel and

each attempt has run into a similar set of problems. The main considerations when building a Minto Wheel are the type of liquid in the tanks, the radius of the wheel, and the heat source, but these can all be changed fairly easily. The following are some improvements to be made in hope's of improving the design of the Minto Wheel. The first is the prevention of a backward rotation of the wheel. The wheel starts from rest and is initially powered by the movement of the liquid from the bottom tanks to the top tanks, but at the beginning, there is no specified direction. This problem may be solved by the application of mechanical brakes or a ratchet to allow rotation only in the specified direction.

Another design flaw that needs improvement is the type of tank that is used. The design of the wheel requires highly pressurized tanks and a temperature gradient to transmit the liquid from one tank to another. Lightweight tanks allow the mass change of the moving liquid to have more of an impact on the rotation of the wheel but often can handle less pressure. This may result in stretching or breaking of the tanks. The design flaws are minimal but hard to eliminate. The wheel's efficiency depends less on the fuel like most mechanical systems and more on the design. When constructed properly, the Minto wheel is a highly efficient way to produce mechanical energy.

Design Narrative

Design approach and evaluations of alternatives:

The largest constraint found when beginning the design process was the temperature difference between the air and the water in the basin. There is a constraint on how large the temperature difference can be between the air and the water because the solar water heater can only heat the water by small amounts with the sun lamps. With a small temperature difference,

liquid will not change phases and move to the top as quickly as imagined. The higher the temperature difference, the easier it will be for the liquid to change phases quickly once the tanks enter the water. The second constraint was using a liquid with low boiling point at realistic pressure for the tank (propane). This was a constraint on the wheel because the liquid is the driving factor behind the spinning of the wheel. The type of liquid plays a large role in how effective and efficient the wheel is. Liquids with lower boiling points are able to change phases quicker along with a smaller temperature difference which will be needed when the tanks enter the water. The final main constraint for the Minto wheel project was meeting the budget of five-hundred dollars. While being a restriction for the project it is also included in the improvement goals for the wheel, by trying to create an affordable model compared to the previous versions. This played a role in how the wheel was designed by adding another factor besides functionality.

Generating multiple solutions:

Before the final design of this project's Minto Wheel was decided there were multiple other solutions considered. The overall function of the wheel and the process were always the same, but the makeup of the wheel along with ways to keep the process functional had several options. At first, the size of the wheel was discussed. If the wheel was too small it wouldn't be able to function properly, and if it was too big it would not fit in the lab space. It was determined that a 2-6 foot diameter wheel would be the ideal range to work with. The base of the wheel will be a bike wheel frame. The other solution for creating the base was constructing the base from scratch using metal pipe, PVC, and wooden boards. However, it was determined for the sake of time and overall structural integrity that a preconstructed bike wheel frame would work best. Along with the base, there was a consideration between building the containers holding the low

boiling liquid or finding a preconstructed one. After discussions, it was decided to use preconstructed containers for the sake of time, safety, and standardization. Inside the tanks there is the low boiling liquid. Several substances could be used in this role such as propane, R12, and R-134a. Propane was decided on based on the easy availability and cheaper price. Another consideration for the solution for this project was how many low boiling liquid containers to use. The final decision was to use 8. Using 6 is possible but left too much space to have the wheel continually spin without getting stuck, and using 10 took up too much space making the wheel hard to work on.

The process of the Minto Wheel requires a heat source for the low boiling liquid. The two solutions discussed were a hot water bath on the bottom of the wheel and a heater/solar heat off the bottom to the side of the wheel applied to the rim. The heater allowed less space to be occupied by the wheel, but overall took more energy and was less efficient than the hot water bath.

Analysis and Synthesis:

The different options in this design were analyzed from the lens of completing the project's goal. Since the goal was to achieve a cost effective working model, the parts and materials used in the design were selected based on their price whenever possible. For instance, propane is not the best or safest working fluid for this project, but it is the only cheap and readily available option so it was used. Furthermore, propane bottles were chosen as tanks because they could withstand the necessary pressures, they would already include some of the working fluid, and they would not require any welding or post manufacturing for use in the design. In the same way, all other design choices were made from the lens of cost, ease of use, and effectiveness in the design.

Making Trade-offs:

Three main decisions had to be made throughout the semester. The type of liquid used for the wheel, the type of tank that would contain the liquid and the pressure needed inside the tanks. To make these decisions, financial and mathematical evaluations were used. Costs of three potential liquids were compared on a per ounce basis. R-12 was \$4.40 per ounce, R-134a was \$0.80 per oz and propane was the cheapest at \$0.44 per ounce. The second factor in need of evaluation was the boiling points of each liquid. Steam tables were found for each, giving a boiling point at a given pressure. The normal pressure inside the propane tanks sits between 100 and 200 psi, so the boiling point would be between 55 and 99 degrees fahrenheit. The second decision that had to be made was how to approach the tanks. The decision between premade versus manually constructed tanks was far simpler than decisions for other considerations. Even though making tanks would allow for more unique specification to the wheel, the negative factors would make it not worth it. Making tanks would add an entirely different section of design and construction that would take focus away from the design of the wheel. With premade propane tanks being available and the correct size, this decision was the obvious choice, especially on our low budget goal for the project as a whole.

Considering Risks:

The two main risks with this project both pertained to the working fluid. First and foremost, propane is highly flammable and can be dangerous. This project required the heating of propane. However, the design could not include any open flames to do the heating as any leak in the tanks would result in an explosion that could severely injure those in the area. Also, pressurizing propane in tanks could lead to an explosion if the tanks themselves could not

withstand the pressure. The pressures needed for the project did not exceed the already known maximum values for the tanks but solid mechanics principals were also applied to ensure that these tanks would be safe to use.

Detailed description of the design:

Below in Table 1, the list of parts necessary to construct the Minto wheel can be seen. This parts list is general and does not include the cost breakdown in this section of the report. The cost of these parts will be further evaluated later in the report.

Table 1: Minto Wheel Design Parts List

Part Description	Number of Units
Full 16 oz propane bottles	4
Empty 16 oz propane bottles	4
4 ft propane hose	4
1" 2ft schedule 40 pipe	2
Metal Ratchet	1
Heavy Base	2
Water container	1
Solar Water Heater	1
Bike Wheel	1
Propane Tank Adapter	4-8
Tank Clamps	8

A major design consideration for the project was the structural integrity of the propane bottles used for the tanks. Since these tanks would be experiencing pressures from the heated fluid, they could fail and cause injury to those in the area. The integrity of the tanks was tested

using a solid work simulation. In this simulation the top quarter of the tank was created and tested. Since the tank is symmetric, if this piece of that tank did not fail then it could be assumed that the rest of the tank will not fail as well. Most of the dimensions of the tank were known. The one value that was not known was the radius of this filet between the main body of the tank and the valve. Since this was not known, a small radius of 0.175 in was used because if this did not fail, then any radius larger than this would not fail as well. The material used for the tank was alloy steel as thus is the material listed for most propane tanks. The pressure the tank was subjected to was the maximum possible pressure dictated by the pressure relief valve on the tank. This valve opens to release pressure when the pressure exceeds 200 psi, so the maximum possible pressure the tank could experience was assumed to be 200 psi. Based on pressure vessel codes, a FOS of 3 is required for vessels to be considered safe to use. For this reason, the FOS between the maximum induced stress and the yield stress of the material was calculated and shown to be greater than 3. This means that the propane tanks will be able to withstand the pressures in the project and are considered safe by engineering standards. Included below are a table of the important values from the simulation as well as an image of the pressure vessel created for the simulation.

Table 2: Pressure vessel simulation test

Yield Strength of Alloy Steel	Max Stress found in Simulation	FOS
90 ksi	28 ksi	3.21

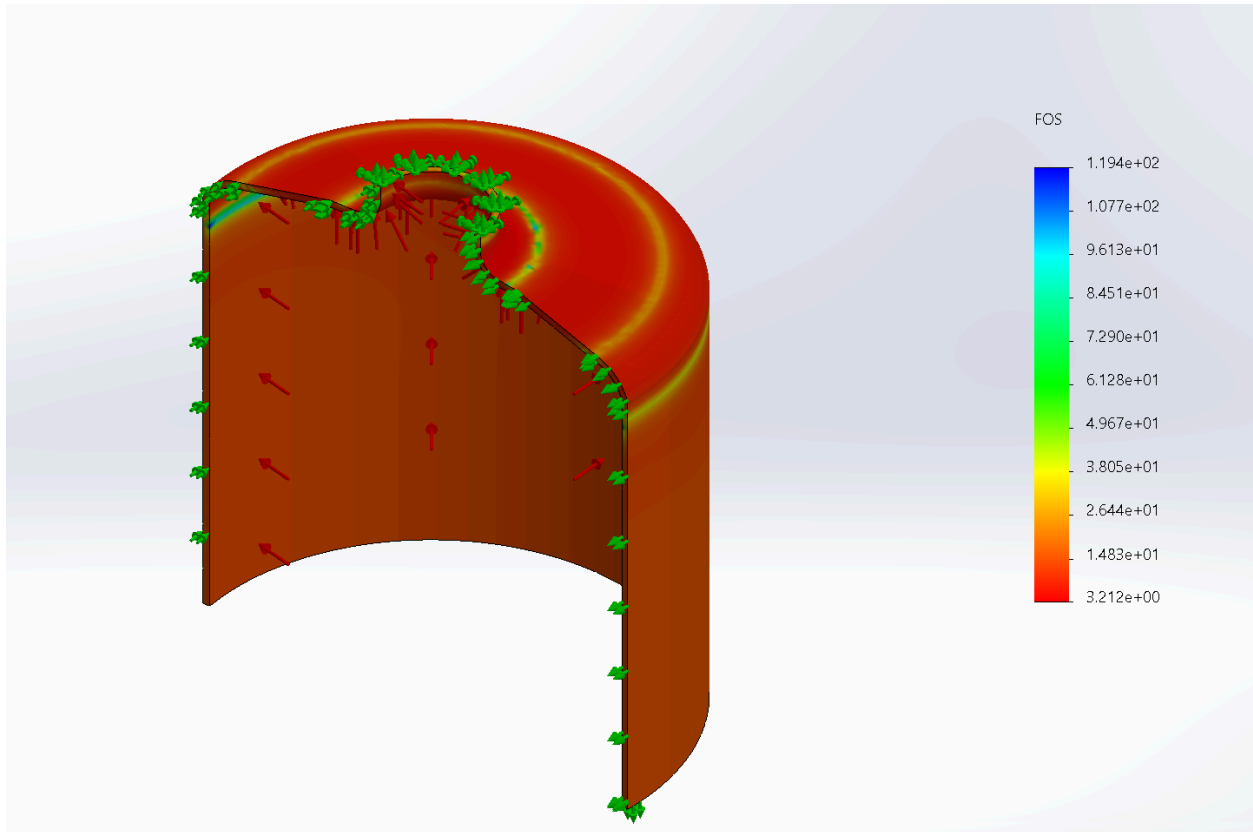


Figure 3: Pressure Vessel Simulation with FOS values

Professional & Societal Concerns, Ethics, Codes, & Standards

Professional and ethical codes and standards are one of the key principles in the field of engineering. These codes and standards not only help keep the general public safe, but also hold engineers to a high degree of accountability. The National Society of Professional Engineers has a Code of Ethics for Engineers that all engineers must adhere to. This code provides a standard in the industry for everyone to follow and use as an example. In the “Professional Obligations”

section in the code of ethics there is a clear statement of “Engineers shall at all times strive to serve the public interest”. (Code of Ethics) This is a crucial part of the idea behind this project. It further goes on to state in a following subsection, “Engineers are encouraged to adhere to the principles of sustainable development in order to protect the environment for future generations.” That is precisely the idea behind the Minto Wheel. The Wheel when running is a constant provider of sustainable, clean energy. This, if applied on a larger scale than this project, could greatly benefit the public need for clean energy. Another standard paramount to this project is also under “Professional Obligations”. It states, “Engineers shall be guided in all their relations by the highest standards of honesty and integrity.” (Code of Ethics) The idea of honesty is arguably the most important engineering principle. The code then continues on to say, “Engineers shall acknowledge their errors and shall not distort or alter the facts.” This is relevant to this project as the group will ensure every communication, transmission of data, conversation, and report will be nothing but truthful and transparent. The main goal of this project is to create a working Minto Wheel with limited resources. If this goal is not realized, there will be no falsification of data or reports to suggest otherwise. Lastly, under “Rules of Practice” the code states “Engineers shall hold paramount the safety, health, and welfare of the public.” (Code of Ethics) This applies to the use of propane as the low boiling liquid in the design. The design of the Minto Wheel relies on pressurized containers holding said low boiling liquid. There are industry safety standards followed for the production of propane tanks. Propane tanks are considered class 1 division 2 pressure vessels. Following the ASME Section VIII code, these pressure vessels are produced with an allowable stress factor of safety of 3. (Hedderman, 2021) Once the pressure in a propane tank reaches 200 psi, a pressure relief valve will be triggered to prevent an explosion of the tank due to too much pressure. (Ferrellgas, 2020) The other concern

with propane is that it is highly combustible. As a result, extreme caution will be taken when dealing with it. Calculations on the proper amount of propane necessary will take place before dealing with the propane. Also, per Virginia fire code 6103.2.1.4 “In Group E and I occupancies, portable LP-gas containers are allowed to be used for research and experimentation. Such containers shall not be used in classrooms.” The group will ensure to wear proper safety gear when dealing with it along with operating in a specific, non-enclosed environment with minimal people and other safety concerns. (Virginia Fire Code, 2012) This will limit the risk involved in dealing with the propane.

Cost and Economic Evaluation

The approximate cost for the parts list can be seen below. These costs are based on the lowest prices found online. At the end of the section an approximate total cost is given in Table 4, but this is subject to change based on the project's needs. Table 2 includes the price differences for various working fluids to show why propane was chosen. However, the actual propane used in the project will be included in the tanks purchased for the wheel so the costs shown in Table 1 are not reflected in the final total cost. Table 3 gives the costs of various parts that we will be considering to construct our design plan. Shipping and tax are estimated for the total cost but not included in each part cost.

As it relates to the selection of materials for the wheel, decisions were made with the project goal in mind. The goal of this project was to create a working prototype of the wheel in the most cost efficient manner possible. The goal of the project was not to create a marketable product. As such, the longevity, maintenance, and aesthetics of the wheel were not considered when making purchasing decisions. These considerations would have likely caused an increase in price, which could be justified to a consumer by the potential tax breaks of using renewable

energy, but were not justifiable for this project as it is not geared towards taking a product to market.

Table 2: The cost of the fluids we will be considering as the medium in the Minto Wheel design.

R-12	R-134a	Propane
\$4.40 oz	\$0.80 oz	\$0.44 oz

Table 3: Individual Parts Cost - See Appendix A for purchasing links.

Part Description	Cost Per Unit	Number of Units	Total Part Cost	Link
Full 16 oz propane bottles	\$5.75 per bottle	8	\$46.00	Bass Pro Shops
4 ft propane hose	\$17.99 per hose	4	\$71.96	Amazon
Propane Tank Adapter	\$10.00	4	\$40.00	Will be purchased after hose and tank connection types have been determined
1" 2ft schedule 40 pipe	\$9.00 per pipe	2	\$18.00	Amazon
Metal Ratchet	\$10.00 per ratchet	1	\$10.00	Waiting until bike wheel purchase
Heavy Base	Free	2	\$0.00	Lab Materials
Water container	Free	1	\$0.00	Lab Materials
Solar Water heater	Free	1	\$0.00	Lab Materials
Bike Wheel	\$90.00	1	\$90.00	Amazon
Tank Clamps	\$6.00	8	\$48.00	Amazon

Table 4: The final total cost that takes into consideration the parts, estimated shipping, and taxes.

Parts Total Cost	\$323.96
Estimated Shipping + Tax	\$150.00
Final Total Cost	\$473.96

Tentative Schedule

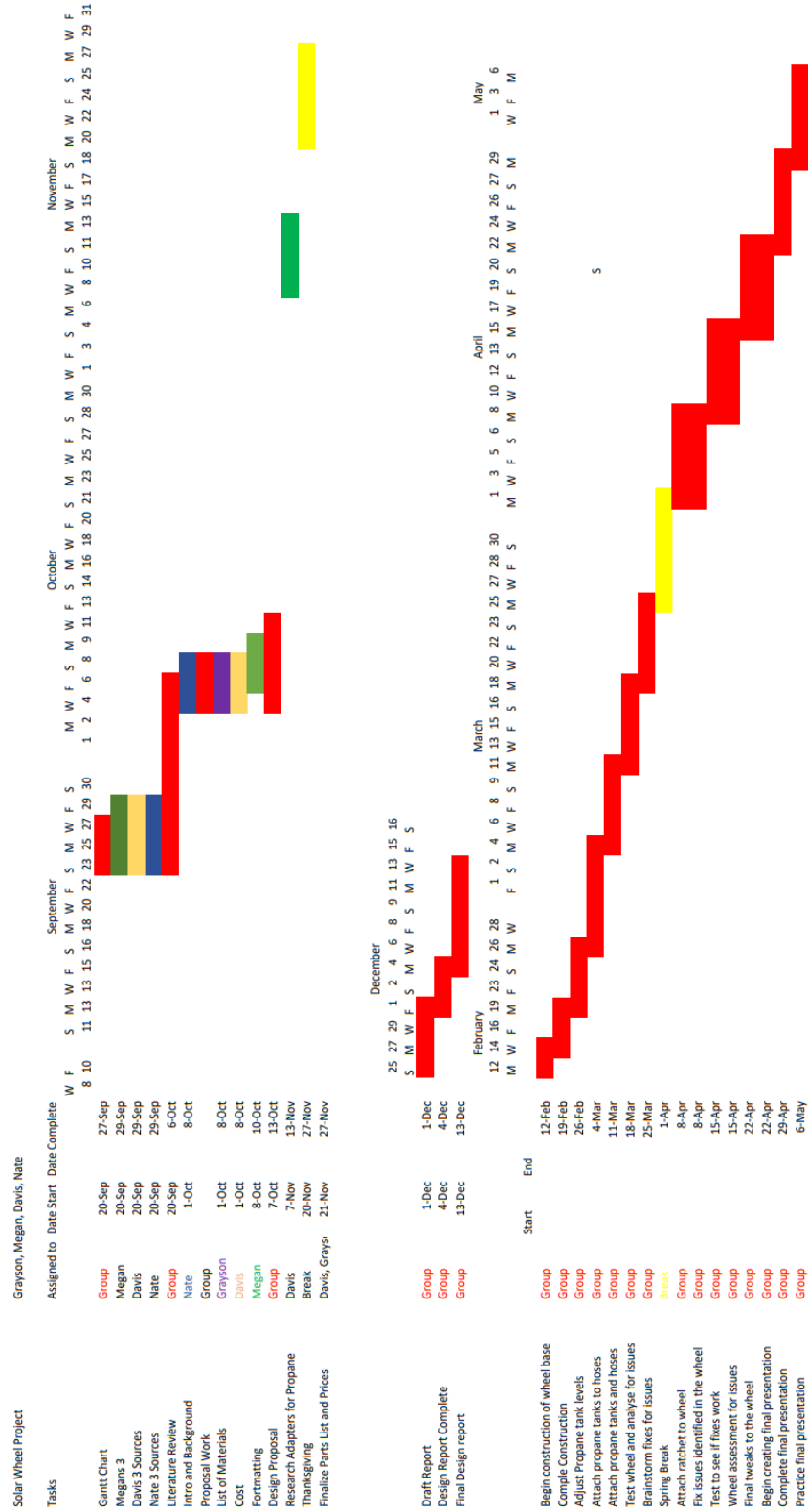
The construction and completion of the Minto wheel will be implemented in phases. The team will meet every Friday and additionally outside of allotted class time to ensure goals are met. The team will arrive prepared to discuss pertinent information and any issues they have encountered. At first, the project will be planned along with the creation of a tentative schedule. Then, the materials needed will be ordered and examined. The wheel will then be constructed by assembling the base, and then adding the extremities on. Once construction is completed, the wheel will be assessed for improvements and final touches. Following the construction of the wheel, a final design report and presentation will be created and presented to all necessary and interested parties. The tentative schedule can be seen in Table 5.

Table 5: The tentative schedule for the Minto Wheel construction.

Week	Tasks To Complete
10/16/23	-Continue to finalize design -Update parts list as needed -Choose which base to use
10/23/23	-Group check in on research and ideas -Decide which base to use -Update parts list -Analysis on propane tanks and necessary pressure

10/30/23	-Examine hot water bath and sizing -Initial completed parts list with pricing -Continued analysis on propane tanks
11/6/23	-Initial completed parts list with pricing -Continued analysis on propane tanks
11/13/23	-Research adaptors needed for propane connection
11/20/23	-Thanksgiving
11/27/23	-Finalize parts list -Finalize parts list prices
12/4/23	-Order parts
2/12/24	-Begin construction of wheel base
2/19/24	-Complete construction of wheel base
2/26/24	-Adjust propane tank levels
3/4/24	-Attach propane tanks to hoses
3/11/24	-Attach propane tanks and hoses to base of wheel
3/18/24	-Test wheel and analyze for issues
3/25/24	-Brainstorm fixes for issues
4/1/24	Spring Break
4/8/24	-Attach ratchet to wheel and test to ensure ratchet is useful -Fix issues identified in the wheel
4/15/24	- Test to see if fixes work -Wheel assessment for issues
4/22/24	-Final tweaks to the wheel -Begin creating final presentation
4/29/24	-Complete final presentation
5/6/24	-Practice final presentation

Related Gantt Chart located below



Summary of Constraints

The following is a table showing a summary of the constraints for the design project and creation of the prototype that have been considered by the group. Beside each constraint is a solution that the group came up with in order to meet the constraints and complete the project within the allotted time frame. This summary of constraints can be seen below in Table 6.

Table 6: Summary of Constraints for the Minto Wheel construction.

Constraint	Application
Accessibility	The project could only use materials that were easily purchased. For example, R-12 could not be used for the working fluid as it can't be purchased by individuals.
Aesthetics	This project did not focus on the aesthetics of the wheel, only its functionality.
Codes	Design and construction needed to be in line with the NSPE code of ethics.
Constructability	The wheel will need to be designed in such a way that complex techniques would not be needed for assembly.(Welding, etc...)
Cost	The project will need to be completed within the \$500 budget.
Ergonomics	Not applicable to this project.
Extensibility	The project's goal is to create a working Minto wheel. Designing in such a way as to use its mechanical energy was not considered.
Functionality	The project would only be successful if the wheel operated as was designed.
Interoperability	Not applicable to this project.
Legal Considerations	This project will need to give credit to any

	sources used to inspire the group's design.
Maintainability	The bike wheel was used as the frame to allow easy access maintenance to the wheel at any time.
Manufacturability	Mass manufacturing would be very difficult without a large factory for all the materials.
Marketability	This product is easily marketable due to its clean sustainable energy and low cost.
Policy	Not applicable to this project.
Regulations	Not applicable to this project.
Schedule	The design proposal and final product needed to be completed by set dates.
Standards	Engineering ethics and standards of safety needed to be considered in the design of the wheel. See "Professional & Societal Concerns, Ethics, Codes, & Standards" section.
Sustainability	The wheel is sustainable as long as the water is heated. That will mostly come from the sun, but artificial heat can also be used.
Usability	The setup and operation of the wheel needed to be designed as simply as possible.

Knowledge, Skills, & Previous Coursework

Throughout this process, knowledge from previous courses was used to influence the design of the wheel. The main area of focus came from thermodynamics. Thermodynamics was used to determine the necessary temperatures and pressures that needed to be achieved in order for the wheel to function. The working fluid needs to be kept as a saturated mixture at all times with the heat provided at the bottom pushing the fluid towards being a saturated vapor, and the

cooling at the top causing the fluid to move towards being a saturated liquid. To do this required calculations which were rooted in the team's knowledge of thermodynamics. Furthermore, solid mechanics was used to determine any possible safety risk with the pressurized tanks. Since the pressures required to make the wheel work were known, solid mechanics equations could be used to determine if the tanks would approach or even exceed failure at these pressures. Finally, the team used CAD experience from the engineering design course work to create a model of the wheel for visualization before building the physical wheel.

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Appendix:

Appendix A: Cost of Parts with order link

Part Description	Cost Per Unit	Number of Units	Total Part Cost	Link
Full 16 oz propane bottles	\$5.75 per bottle	8	\$22.99	https://www.basspro.com/shop/en/coleman-1-lb-propane-cylinder-4-pack?hvarAID=shopping_googleproductextensions&ds_e=GOOGLE&ds_c=BP%7CShopping%7CPCMax%7CCamping%7CGeneral%7CNAud%7CNVol%7CNMT&gclid=CjwKCAiAxreqBhAxEiwAfGfndACU0pj7Q5mRFDK7OKVP9U-brlSyzGw0QrU0BzLtQW62J5NIYXHYdBoCElcQAvD_BwE&gclid=aw.ds
4 ft propane hose	\$17.99 per hose	4	\$71.96	https://www.amazon.com/DOZYANT-Distribution-Extension-Assembly-Throwaway/dp/B06Y678C2P/ref=sr_1_20?keywords=4%2Bft%2Bpropane%2Bhose%2Bwith%2Bmale%2Band%2Bfemale%2Bconnections&qid=170240

				0227&sr=8-20&th=1
Propane Tank Adapter	\$10.00	4-8	\$40.00-80.00	Will be purchased after hose and tank connection types have been determined
1" 2ft schedule 40 pipe	\$9.00 per pipe	2	\$18.00	https://www.amazon.com/Pipe-Schedule-40-White-Custom-Length/dp/B09GL45K6J/ref=asc_df_B09GL2K8CR/?tag=hyprod-20&linkCode=df0&hvadid=598399648009&hvpos=&hvnetw=g&hvrnd=1564398683563402666&hvpon=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvllocint=&hvlocphy=9008375&hvtargid=pla-1840762375377&mcid=e0deb993c4ac35898c2b60a767e652d8&th=1
Metal Ratchet	\$10.00 per ratchet	1	\$10.00	Waiting until bike wheel purchase
Heavy Base	Free	2	\$0.00	Lab Materials
Water container	Free	1	\$0.00	Lab Materials
Solar Water heater	Free	1	\$0.00	Lab Materials
Bike Wheel	\$90.00	1	\$90.00	https://www.ama

				zon.com/HinLlaDa-29-Mountain-Bike-Wheelset/dp/B0CCJFFTJN?source=ps-sl-shoppingads-lpcontext&ref=fplfs&smid=A201G1ZM79RV2T&th=1&psc=1
Tank Clamps	\$6.00	8	\$48.00	https://www.amazon.com/Koehler-Enterprises-KE28BX-Piece-Clamp/dp/B00XAK7NE8/ref=asc_df_B00XAK7NE8/?tag=&linkCode=df0&hvadid=312522113990&hvpos=&hvnetw=g&hvrnd=16799135637354031760&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvloophy=9008375&hvtargetid=pla-625315312527&ref=&adgrpid=64911485929&th=1